



UNIT PURPOSE

Students build a coherent model of matter by connecting macroscopic observations with particle representations, composition, physical and chemical properties, change evidence, and property-based separation methods.

INSTRUCTIONAL RHYTHM

Engage -> Teach -> Model -> Guided Practice -> Independent Practice -> Exit Ticket -> Mixed Review -> Unit Test

SUGGESTED PACING

- 1 Lesson 1: Matter at Two Scales. Use the particle visual to separate observable evidence from model assumptions.
- 2 Lesson 2: Classify Matter by Composition. Require evidence for every label.
- 3 Lesson 3: Properties Describe Matter. Contrast property with change and intensive with extensive.
- 4 Lesson 4: Physical or Chemical Change. Treat common clues as evidence, not automatic proof.
- 5 Lesson 5: Separate Mixtures by Properties. Link each method to the property difference it exploits.
- 6 Cumulative mixed review, feedback, and targeted reteaching.
- 7 Unit Test A. Reserve Form B for a retake, alternate class, or test security.

PACKAGE MAP

Projectable slides teach the lesson. Student pages provide guided notes, modeled thinking, supported practice, independent practice, and exit tickets. Tests A and B follow all instruction and review. The teacher file contains every answer and rationale.



Support, monitor, and respond

BEFORE TEACHING

Preview the particle-model conventions and the documented instructional visuals. Decide whether students will work individually or discuss the engage prompts in pairs. Emphasize that a model is useful evidence but is not a literal, to-scale photograph.

FORMATIVE LOOK-FORS

Students distinguish uniform from pure, property from change, physical clue from chemical proof, and a separation method from the property difference that makes it work.

SUPPORT

Keep the composition decision path and method-property bank visible. Let students annotate particle types and regions. Provide sentence frames that require claim, evidence, and reasoning.

EXTEND

Ask students to propose an additional test that could challenge a classification, compare two viable separation sequences, or identify a limitation in a particle representation.

ASSESSMENT USE

Use Form A after instruction and mixed review. Reserve Form B for retakes, an alternate class period, or test security. Do not use either test as the first exposure to a skill.

SCOPE

Supports the NGSS practices of Developing and Using Models and Analyzing and Interpreting Data; no formal performance-expectation alignment is claimed.